Comparison of Soxhlet and Direct Ultrasonic Techniques for Determination of Polycyclic Aromatic Hydrocarbons in Agricultural and Urban Soils of Pakistan

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Summary: Polycyclic aromatic hydrocarbons (PAHs) pose a potential danger to the environment due to their complex structures and persistent nature. In the present study, two extraction techniques were compared on the basis of capital cost, time and extraction efficiency. Four PAHs (naphthalene, acenaphthene, phenanthrene and anthracene) were extracted by Soxhlet extraction and direct ultrasonic technique from the agricultural and urban soil samples of refinery adjacent area in Rawalpindi district of Pakistan. Gas chromatographic analysis showed that the agricultural soils close to the urban area were severely contaminated by PAHs and concentration of PAHs up to 4198 µg/L was detected in the agricultural soil samples. The results indicated that recovery of PAHs is dependent on the extraction technique and the extraction efficiency by the Soxhlet method was 2-times greater than ultrasonic technique. However, the earlier technique required high solvent volume and time. The results suggested that the preferred technique in terms of extraction recoveries was Soxhlet and more reliable results can be achieved using this method.

Key words: Soil, polycyclic aromatic hydrocarbons, Soxhlet extraction, ultrasonic extraction.

Introduction

Polycyclic aromatic hydrocarbons are a group of more than 100 multiple benzenic ring compounds with ubiquitous nature and 16 PAHs have been identified as priority pollutants by the US Environment Protection Agency [1]. These PAHs are widely studied because of their toxicities, potential of human exposure, carcinogenity and occurrence as a light non-aqueous phase liquids [2]. The 16 PAHs were combined into one profile to avoid repetition across multiple profiles on the individual PAHs since these chemicals often occur together and they are of serious environmental concern [3, 4]. Because of the multiple aromatic ring systems in PAHs, these compounds can absorb light energy in the UVA and for some in the visible range [5], forming reactive species and causing damages to human cellular components. PAHs are relatively neutral and stable molecules. They have low solubility [6] and low volatilities. Their lipophilicity is high as measured by water-octanol partition coefficients (Kow).

The major PAHs source is incomplete combustion of organic materials [7-9]. This occurs naturally in vegetation fire and volcanic activities. Moreover, most PAHs in urban areas are produced by anthropogenic combustion of wood and fossil fuels. Other sources of PAHs includes oil spill accidents, gas flaring and refining operations and wastewater discharge from domestic and industrial activities [10]

For PAHs analysis, effective extraction is most important step. The aim of extraction techniques is to remove the analyte from its matrix effectively. The selection of extraction technique is based upon its capital and operating cost, minimal solvent usage and time efficiency. The range of techniques available makes selection difficult. In the present study, two cost effective techniques were compared for the effective extraction of PAHs.

Several scientists have used ultrasonic agitation method for the extraction of PAHs from soil samples [11-13]: however, its effectiveness has not yet been compared with the Soxhlet extraction method. Nonetheless, Soxhlet, extraction is considered a preferred technique in terms of reproducibility and yield [14].

Understanding the distribution of PAHs in soil is the basis of soil remediation before land reuse. In Pakistan with the rapid growth of population and spread of urbanization, many chemical plants are present in the close vicinity of city. The contamination of these areas poses a significant risk to environmental and human health. Before planning remedial action monitoring the level of contamination is an important step, so present study reports the concentration of selected low molecular weight PAHs in different soils and effectiveness of two extraction techniques was compared for the

maximum recovery of PAHs from agricultural and urban soil samples.

Results and Discussion

The data on physical and chemical properties clearly show that the studied soil differed in their properties (Fig. 1(a-c). Soil pH values were in alkaline range (pH 8-9). Physicochemical parameters of pH, EC and TOC in commercial soils were found quite high.

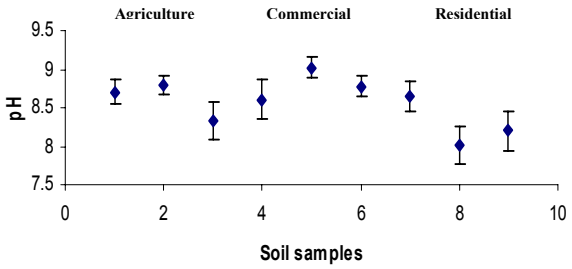


Fig. 1a: pH of soil samples.

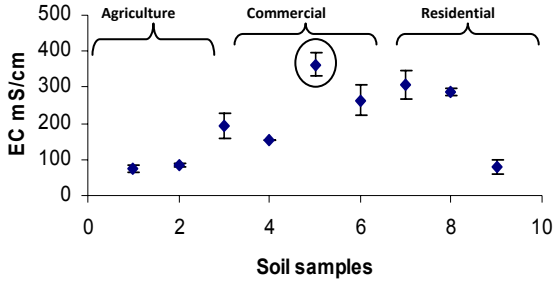


Fig. 1b: Electrical conductivity of soil samples.

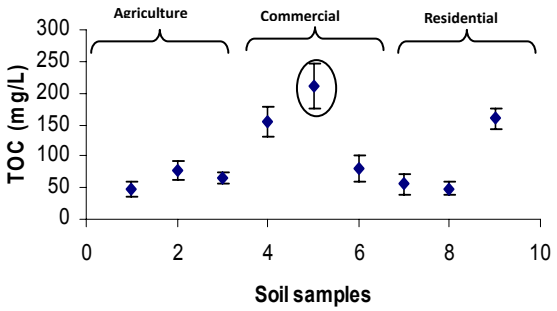


Fig. 1c: Total organic content of soil samples.

The PAHs concentration ranged between 2546-4288 µg/g (average: 3464±592 µg/g) with Soxhlet extraction and 2632-2238 µg/g (average: 1892±640 µg/g) with ultrasonic extraction. Phenanthrene and anthracene were among the highest found PAHs. The highest average concentration of PAHs was found in agricultural soils. Agriculture soils were taken from fellow lands before sowing of next season PAHs attributed to surface deposition. The results for each extraction technique are presented as sum of four PAHs in Fig. 2. Each sample of the contaminated soil was extracted three times and analyzed using GC-FID. In each sample three PAHs i.e. anthracene, phenanthrene and acenaphthene were identified but naphthalene was below the detection limits in 80% samples.

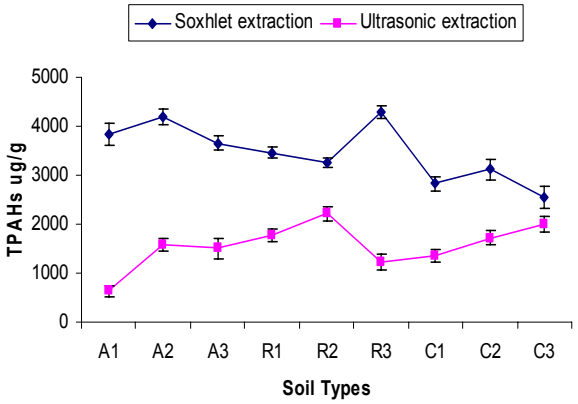


Fig. 2: Total PAHs from Soxhlet and ultrasonic extraction technique.

Statistical significance was determined at the 95% confidence interval as a p-value > 1.860. The correlation of four PAHs in soil is presented in Fig. 3. Results of Pearson correlation matrix suggested a strong positive relation between acenaphthene and phenanthrene with each other.

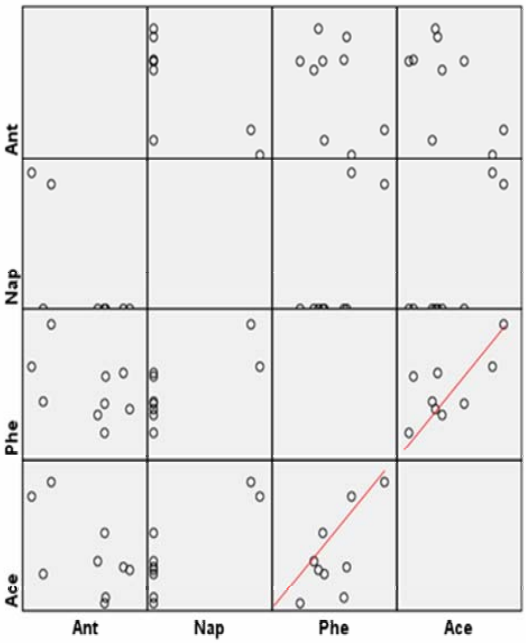


Fig. 3: Pearson correlation matrix for soil samples.

In addition to extraction techniques it was interesting to compare the relative benefits of each extraction technique like capital cost, extraction time, organic extraction solvent volume, and sample weight. The relative merits of extraction technique followed the order as: capital cost Soxhlet> ultrasonic, extraction time Soxhlet (24 h) > ultrasonic (30 min), sample weight Soxhlet (5g) = ultrasonic (5 g) and solvent volume Soxhlet (300ml n-hexane) >ultrasonic (25 ml n-hexane). Desirable features of economic benefit are low cost, minimum solvent volumes and fast extraction time but above all effectiveness of extraction and standard methods is quality benefits.

This study reports concentration of the selected PAHs in agricultural and urban soils, and comparison of two extraction techniques. All the selected PAHs were found in top soil samples as the soil is ultimate residence of atmospheric PAHs. High molecular weight PAHs are more prevalent than low molecular weight PAHs as also observed by [15]. Li et al. [16] also found more PAHs in top soils which supports our findings. The highest average PAHs concentration was observed in a road side site surrounded by tall buildings. Probably, emitted particulate associated PAHs could not get any chance of dispersion but settled in the immediate vicinity. Frequent change of gear and slow speed traffic in the residential areas also results in highest PAHs emissions [8]. Lowest concentration observed in sample C3 (a commercial site nears a play ground) where green grass cover may adsorb maximum PAHs.

While comparing the extraction techniques, the desirable features such as low cost, minimum solvent volume and fast extraction time and above all availability of standard method like US EPA [17] for Soxhlet and extraction efficiency were considered. Soxhlet extraction technique was proved to be better in availability of method and extraction efficiency though its capital cost, extraction time and solvent volume consumption was high. However, its results are considered accurate and reliable than any other technique [18]. There is a possibility that some other extraction technique may challenge Soxhlet in near future.

According to statistical results, PAHs compounds acenaphthene and phenanthrene showed a direct linear correlation in soil samples.

Physicochemical parameters showed a relatively high concentration in commercial soils. It may be due to rapid shuffling of top soil cover in commercial area, too much activity on two adjoining roads. The total organic carbon was quite low in agricultural soil as compared to commercial soil. The commercial soil may be getting high carbon content from multiple sources and it was in the center of urbanized zone.

Experimental

Study Area

Soil samples were collected from Morgah (an area adjacent to oil refinery), Rawalpindi. The soil of this area was analyzed for PAHs.

Soil Preparation

Surface soil samples (0-15 cm upper soil layer) were collected from nine locations (Fig. 4) and homogenized to make a composite sample. Samples were transported to laboratory in an ice box and air-dried for 24 h. Then, the samples were oven-dried at 60 °C for moisture removal and sieved through 2 mm sieve. The samples were preserved at zero Celsius in dark for subsequent studies. Soil preparation was done following [19].

Standard and Solvents

Standard solutions of PAHs i.e., acenaphthene, anthracene, naphthalene and phenanthrene were purchased from Chiron Chemicals, Australia. Standard solutions of PAHs were prepared by serial dilution of the commercial products with n-hexane (GC grade) and stored in glass bottles at room temperature.

Soxhlet Extraction

Soxhlet extractions were performed by using 5 g soil to which 5 g anhydrous sodium sulfate was added. The mixture was transferred into cellulose extraction thimble and inserted into Soxhlet assembly (Gerhardt EV 16, Germany) fitted with 500 ml flask. A 300 ml of n-hexane was added and heated for 24 h using isomental. The extracts were concentrated to 1 ml using a rotary evaporator.

Ultrasonic Extraction

Soil samples were extracted in triplicates by ultrasonic extraction (JAC Ultrasonic 1505) using 5 soil with 25 ml of n-hexane for 30 min at high mode sonication at 25 °C in water bath. Extracts were fractioned by a silica gel column. Silica gel purchased from Merck Darmstadt, Germany was activated at 150 °C for 24 h and kept overnight in desiccators. The extract was eluted through silica gel column and concentrated to 1 ml using rotary evaporator.



Fig. 4: Map representing soil sampling points (A= agricultural soil, R= residential soil and C= commercial soil).

Sample Analysis

Gas chromatographic separation and identification of PAHs was performed on GC (Shimadzu 2010, Japan) with flame ionization detector. A 30 m × 0.32 mm I.D., 0.50 µm film thickness TRB-5 capillary column (Tecknokroma) was used to achieve separation with the following temperature programme: initial column temperature, 150 °C; hold time, 4 min; increase at 10 °C min^{-1} to 280 °C hold for 5 min. The detector temperature was set at 280 °C. The chromatogram of selected PAHs is represented in Fig. 5.

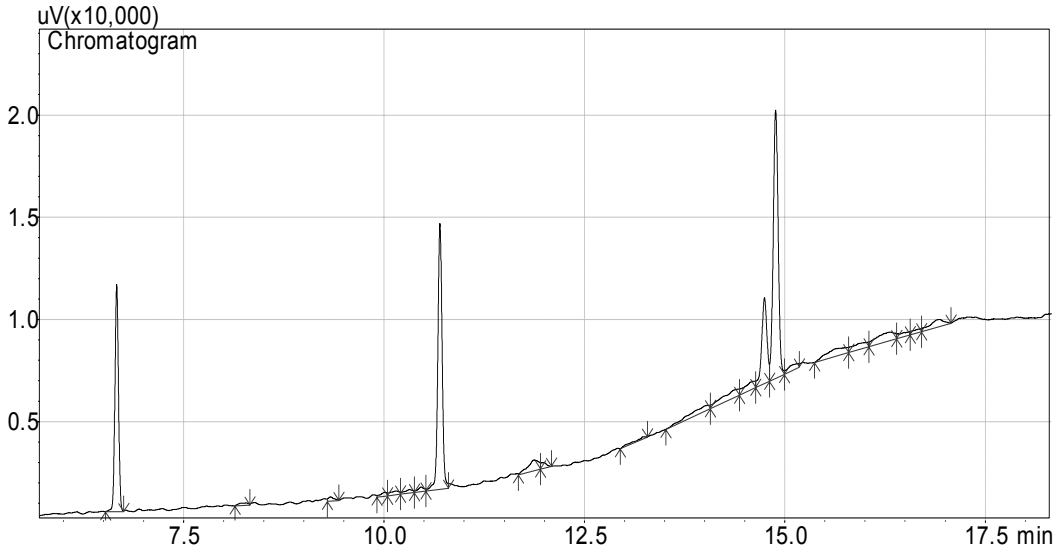


Fig. 5: Chromatogram representing the peak signal and retention times of mixture of PAHs [(a) naphthalene (b) acenaphthalene (c) phenanthrene (d) anthracene].

Total organic content (TOC) of soil samples were determined using TOC analyzer (Analytic Jena). Soil pH and electrical conductivity (EC) were measured using 10 g air dried soil in 100 ml deionized water, using pH meter (600 pH Milwaukee) and EC meter (Hach Sension 5), respectively. All the extraction techniques were operated under standard and optimized conditions determined in the laboratory for PAHs based on manufacturer's recommendations.

Quality Assurance

The quality control program was carried out. This program includes the following: pre-extraction of thimbles and silica with solvent. Blank was run with each set of analysis. Quantification of PAHs were carried out using external standards with coefficient of calibration curves higher than 0.986. A random sample was run in triplicate.

Statistical Analysis

Statistical treatment of the data was done for the selection on ∑PAHs reported. Statistical significance was determined by paired t-test approach [18] at the 95% confidence interval. To investigate the influence of one particular polynuclear aromatic hydrocarbon on the other, correlation was performed between PAHs in soil and water by SPSS software version (16.0).

Conclusion

The present study shows a significant difference of concentration by two extraction techniques. Soxhlet technique gave almost the double extracted concentration than ultrasonic despite its high solvent volume consumption and extraction time. The study suggests that the agricultural soil is under a threat of anthropogenic contamination of petroleum related compounds in urban districts. The investigation of polycyclic aromatic hydrocarbons from soil revealed that the highest concentration of Σ PAHs in soil was 4288 µg/g and it was also very high when compared with Dutch soil standards 0.02-0.05 µg/g and Polish standards 16 0.2-10 µg/g. Our results suggest Soxhlet a better extraction technique but there is still need to evaluate this technique for time saving.

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